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SPECIAL REPORT

EPIDEMIOLOGY, CLINICAL PRACTICE AND HEALTH

Respiratory sarcopenia: A position paper by four professional organizations

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We defined respiratory sarcopenia as a coexistence of respiratory muscle weakness and decreased respiratory muscle mass. Although respiratory muscle function is indispensable for life support, its evaluation has not been included in the regular assessment of respiratory function or adequately evaluated in clinical practice. Considering this situation, we prepared a position paper outlining basic knowledge, diagnostic and assessment methods, mechanisms, involvement in respiratory diseases, intervention and treatment methods, and future perspectives on respiratory sarcopenia, and summarized the current consensus on respiratory sarcopenia. Respiratory sarcopenia is diagnosed when respiratory muscle weakness and decreased respiratory muscle mass are observed. If respiratory muscle mass is difficult to measure, we can use appendicular skeletal muscle mass as a surrogate. Probable respiratory sarcopenia is defined when respiratory muscle weakness and decreased appendicular skeletal muscle mass are observed. If only respiratory muscle strength is decreased without a decrease in respiratory function, the patient is diagnosed with possible respiratory sarcopenia. Respiratory muscle strength is assessed using maximum inspiratory pressure and maximum expiratory pressure. Ultrasonography and computed tomography are commonly used to assess respiratory muscle mass; however, there are insufficient data to propose the cutoff values for defining decreased respiratory muscle mass. It was jointly prepared by the representative authors and authorized by the Japanese Society for Respiratory Care and Rehabilitation, Japanese Association on Sarcopenia and Frailty, Japanese Society of Respiratory Physical Therapy and Japanese Association of Rehabilitation Nutrition. **Geriatr Gerontol Int 2023; 23: 5–15**.

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Introduction

Sarcopenia is the generalized loss of skeletal muscle mass and muscle function that occurs with aging,[1] but skeletal muscle changes do not necessarily occur throughout the body. Breathing is a vital function performed by respiratory muscles, and when respiratory muscle function is impaired, serious problems occur, including decreased ventilatory reserve. This leads to dyspnea on exertion, exercise intolerance and reduced physical activity, which in turn worsens the prognosis.

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The respiratory muscles, including the diaphragm and various accessory respiratory muscles, work in coordination to fulfill respiration. During resting respiration, the diaphragm acts almost exclusively as the inspiratory muscle, whereas expiration is accomplished by the elastic contractility of the chest cage. Specific maneuvers are required to evaluate respiratory muscle strength, and it cannot be evaluated at rest. Advances in imaging techniques have made it possible to evaluate respiratory muscle mass quantitatively and qualitatively, and quantitative image analysis contributes to understanding pathophysiology by local information on each respiratory muscle.

Unfortunately, respiratory muscle function assessment is not incorporated into the usual pulmonary function tests. Meanwhile, respiratory muscle dysfunction is an essential determinant of physical function and ventilatory dysfunction. Moreover, a specific intervention for respiratory muscle, such as inspiratory muscle training (IMT), represents a close relationship between evaluation and therapeutic intervention on respiratory muscle strength. Therefore, respiratory muscle strength should be evaluated separately. In addition, etiologic factors include aging, physical inactivity, malnutrition, lung diseases, cancer-related cachexia, iatrogenesis (surgical and pharmacological intervention) and neuromuscular diseases should be noted.

Respiratory sarcopenia is defined as a condition in which there is both low respiratory muscle strength and low respiratory muscle mass. We chose this definition because low respiratory muscle strength is essential in respiratory sarcopenia, just as muscle strength is becoming more critical in whole-body sarcopenia, and because respiratory dysfunction is often observed in lung diseases other than respiratory sarcopenia. We developed this position paper because a unified view on the involvement and mechanism of each respiratory disease, diagnosis, treatment, clinical significance and outcome, and future perspectives is needed. The board of directors of the four societies agreed to prepare this paper, and the representative authors from those societies wrote the paper, which was approved by the board of directors.

Role and characteristics of respiratory muscles

Respiratory muscles consist of expiratory and inspiratory muscles. In addition to individual characteristics such as abdominal and thoracic breathing, respiratory muscles are responsible for various breathing patterns such as upper thoracic breathing, mandibular breathing and other abnormal breathing patterns, as well as ventilatory movements to sustain exercise performance and life. Expiratory muscles are the abdominal wall muscles, including the internal intercostal muscles, transversus abdominis, internal and external oblique abdominal muscles, and rectus abdominis. Inspiratory muscles are mainly the diaphragm and external intercostal muscles. During resting expiration, expiration is usually guided by the elastic forces of the chest cage and lungs themselves. The expiratory muscles participate during effort breathing to increase intrathoracic pressure, increase expiratory flow and obtain ventilatory volume, including coughing and sneezing, which are responsible for eliminating excess secretions and foreign substances. It also plays an essential role in inspiration by controlling the muscle length and tension of the diaphragm through its action of compressing the viscera and pushing up the diaphragm. The expiratory muscle strength exerted by the abdominal expiratory muscles affects coughing power.[2] Diaphragmatic contraction plays a major role in most (approximately 80%) of inspiratory movements at rest. The diaphragm also has physiological functions besides respiratory function, stabilizing posture by regulating intra-abdominal pressure, assisting in defecation and delivery, and participating in circulation, such as cardiac function and lymph.[3]

Muscle fibers of the diaphragm are composed of type I fibers (slow twitch muscle) and type II fibers (fast twitch muscle). Levine *et al.* reported by using muscle biopsy that type I fibers were more prevalent in severe chronic obstructive pulmonary disease (COPD) (71%) than in the control group (42%).[4] This change results from adaptation to increased fatigue resistance in severe COPD.[5] Fatigue of respiratory muscles leads to fatigue-induced metabolism that increases sympathetic vasoconstrictor outflow, causing reduced blood flow to appendicular muscles and muscle fatigue.[6] In addition, age-related loss of chest wall flexibility increases the load on respiratory muscles. This increases the demand for blood drainage from the heart to the respiratory muscles, resulting in reduced blood flow to the appendicular muscles.[7]

Indices of respiratory muscle function are therapeutic characteristics in various clinical settings. Diaphragm thickness (Tdi) change[8] and mobility[9] are used as reference values for extubation in ventilated patients in the intensive care unit (ICU). Respiratory muscle strength is one of the functional outcomes of respiratory rehabilitation[10] and may contribute to improving exercise tolerance and dyspnea in patients with COPD.[11] Furthermore, diaphragm mobility is involved in improving exercise tolerance,[12] supporting the possibility that respiratory muscle functional measures may contribute as prognostic factors.

Evaluation of respiratory muscles

Voluntary tests of respiratory muscle strength

The main indices used in clinical practice are listed below. Each index has its characteristics regarding specificity, accuracy, simplicity, reproducibility, availability and cost (Table 1).

Maximal static expiratory and inspiratory mouth pressure

Maximal expiratory pressure (MEP) and maximal inspiratory pressure (MIP) are currently the most widely used indices in clinical practice. However, specialized equipment is required for their measurement. Static MEP and MIP are measured to assess overall respiratory muscle strength.[13] MEP is performed at the maximum expiratory effort at the total lung capacity level, and MIP is usually performed at the maximum inspiratory effort at the residual volume level. The maximum value is recorded when the difference is less than 10% in the three measurements.

Average values for over 65 years of age were MEP 116 cmH_2O and MIP 57 cmH_2O for women, and MEP 174 cmH_2O and MIP 83 cmH_2O for men.[14] However, the range varies among studies. It also depends on the model and the shape of the mouthpiece.[13]

Peak expiratory flow rate

Peak expiratory flow rate (PEFR) is the maximum value of expiratory flow rate during the measurement of forced vital capacity (FVC). It is the height of the apex of the flow volume curve. After resting breaths, the subject inspires from the resting expiratory level to the maximal inspiratory level and then exhales to the maximal expiratory level by exhaling effort at maximal force. The normal PEFR is 450–550 L/min for adult men and 320–470 L/min for adult women, and these values decline with age. Three consecutive expulsions at maximum effort are made, and the maximum value is taken as the measured value. Recently, an electronic type is also available.

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Those with whole-body sarcopenia have lower PEFR than those without.[15] Lower PEFR is associated with lower muscle mass,[16] which is useful for determining whole-body sarcopenia in community-dwelling older people.[17]

Peak cough flow

Peak cough flow (PCF) is an effective predictor of sputum expectoration. PCF can be easily measured with a peak flow meter and a face mask or mouthpiece. The measurement is performed in sitting, with the mask or mouthpiece connected to a pneumotachograph or peak flow meter. The subject is asked to cough after maximal inhalation, and the peak flow at that time is used as the PCF. Three to six measurements (<5% variability) are performed, and the maximum value is recorded as the PCF. The PCF of a normal adult is approximately 470–600 L/min,[18] and this value declines with age.

Maximal sniff nasal inspiratory pressure

Maximal sniff nasal inspiratory pressure (SNIP) is measured by a pressure transducer connected to a catheter placed in the nasal cavity. Inhale through the nasal cavity strongly and early at the functional residual volume level, and the peak pressure is the measured value. SNIP is highly correlated with FVC and MIP,[19] and is associated with diaphragmatic activity.[20] SNIP has also been validated in COPD as well as in healthy individuals,[21] and healthy children and healthy adults show similar values.[22] Normal values of SNIP are 104 ± 26 cmH_2O in men and 93 ± 23 cmH_2O in women and are related to age, sex, weight and height.[22] However, SNIP is not widely used in clinical practice due to the limited number of models available for measurement in Japan.

Ultrasonography

Studies on diaphragm evaluation using ultrasound diagnostic equipment have been conducted in many areas and are highly useful and accurate in diaphragm assessment. Currently, the most commonly used methods are (i) Tdi, (ii) Tdi change with contraction (thickening rate) and (iii) diaphragm mobility with respiration, and sometimes muscle brightness is evaluated.[23] The diagnostic value is diminished with ultrasound diagnostic equipment that does not have sufficient resolution. Moreover, accurate evaluation requires the training of the evaluator in diaphragm measurement.

Diaphragm thickness (Tdi). Tdi is a static measurement of Tdi. It is usually measured at the end of expiration. Tdi measurements in healthy people are highly reproducible, and ultrasound Tdi correlates well with directly anatomically measured Tdi.[24] The lower limit of Tdi in healthy adults is 0.15 cm, but there is a range of reference values.[25] Therefore, it is not clear whether a Tdi below this threshold is diagnostic of diaphragmatic dysfunction. Tdi is related to factors such as age, body mass index and gender, and it must consider subject characteristics.[26] For example, if the diaphragm is elevated due to low diaphragm strength, the muscle thickness becomes thinner as the muscle length increases. In contrast, if the diaphragm is flattening due to COPD or other conditions, the muscle thickness becomes thicker as the muscle length becomes shorter.

Diaphragm thickening fraction and ratio. Tdi is measured dynamically using M-mode, and the ratio of Tdi in inspiration and expiration is the thickening ratio or thickening fraction. The thickening ratio or thickening fraction is reproducible, and the lower limit of normal during effort breathing in healthy people and COPD is 20%.[27]

Diaphragm excursion. Diaphragm excursion is measured by the diaphragm movement distance using M-mode.[28] There are left–right differences in the excursion, with high reproducibility of mobility of the right diaphragm, with a lower limit of 3.6 cm for women and 4.7 cm for men during maximal inspiratory effort in healthy people.[28] Diaphragm excursion is sensitive to changes in breathing patterns and is used to differentiate diaphragm weakening after exacerbations in patients with COPD.[29]

Other respiratory muscle assessment methods

Computed tomography (CT) images are used to evaluate respiratory muscles because of their superior resolution despite the radiation exposure problem. It is challenging to evaluate respiratory muscles, specifically with magnetic resonance imaging and dual-energy X-ray absorption, which are used to diagnose whole-body sarcopenia. Because of the frequent use of chest CT in evaluating respiratory diseases, several post-processing evaluation methods have been reported. Evaluation items include muscle thickness, cross-sectional area and three-dimensional shape. Respiratory muscles that can be evaluated include the diaphragm,[30,31] intercostal muscles[32] and respiratory support muscles such as the pectoralis major and minor muscles[33] and erector spinae.[34] However, reference values for the muscle mass of these muscles, including the scalene and sternocleidomastoid muscle, which are representative accessory respiratory muscles, have yet to be reported.

Aging, sarcopenia and respiratory sarcopenia

Skeletal muscle mass decreases with age, with functional declines such as walking speed and grip strength. The European Working Group on Sarcopenia in Older People published revised

Table 1 Summary of the characteristics in voluntary tests of respiratory muscle strength

Tests	Specificity[†]	Accuracy	Simplicity	Reproducibility	Availability	Cost
Maximal mouth pressure	+++	++	++	++	++	High
PEFR	+	++	+++	++	++/+++[†]	High/low[†]
PCF	+	++	+++	++	+++	Low
SNIP	+	++	+++	++	+	High
Ultrasound	+++	+	+	+	++	High

PCF, peak cough flow; PEFR, peak expiratory flow rate; SNIP, maximal sniff nasal inspiratory pressure.

[†]Whether both inspiratory and expiratory muscle strength components can be measured.

Use of spirometer/peak flow meter.

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diagnostic criteria in 2018,[35] followed by the Asian Working Group for Sarcopenia in October 2019.[36] There is now an international consensus that sarcopenia is diagnosed by low muscle mass, strength and physical function.

Sarcopenia is related to various factors, including aging, physical inactivity, malnutrition, endocrine changes (hormones, cytokines and humoral factors) and abnormal neuromuscular junction, and is greatly influenced by comorbidities. Decreased synthesis and increased degradation of muscle proteins, autophagy failure, mitochondrial dysfunction and impaired muscle repair have been considered as mechanisms. The relationship between the renin–angiotensin system, myokines, muscle microcirculation and sarcopenia is also being clarified.[37]

Cellular changes in the muscle of sarcopenia include a decrease in the size and number of muscle fibers, with type II fibers particularly affected. In addition, the age-related transition of muscle fibers from type II to type I, fatty infiltration within and between muscles (myosteatosis), and a decrease in satellite cells of type II fibers[38] are contributing factors. Molecular changes in muscle involved in sarcopenia include alterations in complex signaling pathways such as insulin growth factor-1, mTOR, forkhead protein transcription factor and other linked pathways.[39]

Muscle strength and muscle mass of respiratory muscles decrease with age. The diaphragm is affected by aging and sarcopenia.[40] Transdiaphragmatic pressure, a measure of diaphragm muscle activity, decreases by 20–41%, and overall respiratory muscle strength decreases by 30% in older people.[40]

Respiratory muscle strength and function are related to whole-body muscle mass, strength and physical capacity. Skeletal muscle mass index correlates with MIP and MEP in older people.[41] Diaphragm weight in autopsy cases of older people is correlated with antemortem lean body mass.[42] Handgrip strength is associated with MIP and MEP in older people[41] and is associated with MIP and PCF in male nursing home residents.[43] MIP is associated with knee extensor strength and grip strength[44] in men and women, including young adults. Moreover, respiratory muscle strength is associated with decreased mobility.[45]

Interaction among whole-body sarcopenia, respiratory sarcopenia and COPD as an example, is shown in Figure 1.

Definition, diagnosis and outcomes of respiratory sarcopenia

Definition and diagnosis

Respiratory sarcopenia is defined as a condition in which there is both low respiratory muscle strength and low respiratory muscle mass. Respiratory muscle strength is currently the most reliable measure of respiratory muscle function in clinical practice. Respiratory muscle mass should be measured when low respiratory muscle strength is observed. Respiratory sarcopenia is confirmed when low respiratory muscle strength and low respiratory muscle mass are both present. If respiratory muscle mass is difficult to measure, we can use appendicular skeletal muscle mass as a surrogate. Probable respiratory sarcopenia is diagnosed when low respiratory muscle strength and low appendicular skeletal muscle mass are observed. Respiratory muscle weakness may be present even without whole-body sarcopenia.[46] Respiratory muscle weakness due to respiratory dysfunction is diagnosed when only respiratory muscle weakness is present and respiratory dysfunction such as restricted or obstructive ventilatory dysfunction is present. If only respiratory muscle strength is decreased and there is no decrease in respiratory function, the patient is considered to have possible respiratory sarcopenia (Figure 2). The assessment of muscle strength and mass for diagnosing respiratory sarcopenia is shown in Table 2.

Outcomes

Respiratory sarcopenia can worsen respiratory function, physical performance, activities of daily living and prognosis. Table 3 summarizes outcomes due to respiratory sarcopenia.

Diseases and respiratory sarcopenia

Chronic obstructive pulmonary disease

Prevalence

Sarcopenia is recognized as an important comorbidity of COPD. With regard to respiratory sarcopenia, in a study evaluating 828 patients with COPD, the %MIP decreased with increasing severity from 87.5% in Stage 1, 87.8% in Stage 2, 78.6% in Stage 3 and 64.3 in Stage 4, while the %MEP was relatively retained.[57] One study evaluated Tdi; however, it was in contrast to normal values and did not evaluate the frequency of respiratory sarcopenia.

Mechanisms

Moreover, both MIP and MEP decline as the disease progresses, particularly as the flattened diaphragm. In addition to diaphragmatic motor dysfunction due to lung hyperinflation, muscle weakness of respiratory muscles other than the diaphragm (inspiratory and expiratory muscles) because of malnutrition causes worsening dyspnea, exercise intolerance and physical inactivity. As such, whole-body and respiratory sarcopenia may be intense pathophysiological features of COPD (Figure 1).

The association with systemic inflammation has been suggested, including an association with endoplasmic reticulum stress, and a different profile has been reported between patients with lung cancer and patients with COPD.[58] In addition, systemic inflammation is recognized as one of the critical mechanisms of other comorbidities of COPD, such as atherosclerosis, osteoporosis and cachexia. These comorbidities may result in physical inactivity and consequently cause disuse atrophy of skeletal muscles, including respiratory muscles.

The flattening of the diaphragm reduces the conversion of tension into pressure according to Laplace's law. Therefore, diaphragmatic "shape" has functional significance, and evaluation of diaphragmatic shape using chest CT and ultrasonography as well as diaphragm muscle mass has certain clinical significance.[30,59] The improvement in respiratory muscle strength achieved by lung volume reduction surgery supports this theory.[60]

Outcomes

Lower MIP is associated with an increased risk of exacerbations.[61] Decreased muscle mass in the accessory respiratory muscles, such as the thoracic and paraspinal (erector spinae) muscles, is associated with poor prognosis,[62] whereas frequent COPD exacerbations cause a decrease in this muscle mass.[63]

There is a strong correlation between exercise tolerance, 6-min walk distance (6MWD) and MIP.[64] However, the associations are not consistent. After 8 weeks of respiratory rehabilitation, there were no significant changes in MIP and MEP, while other indices improved.[65] In addition, an association between improved exercise tolerance and diaphragmic function after rehabilitation has been shown in a study from Japan.[12]

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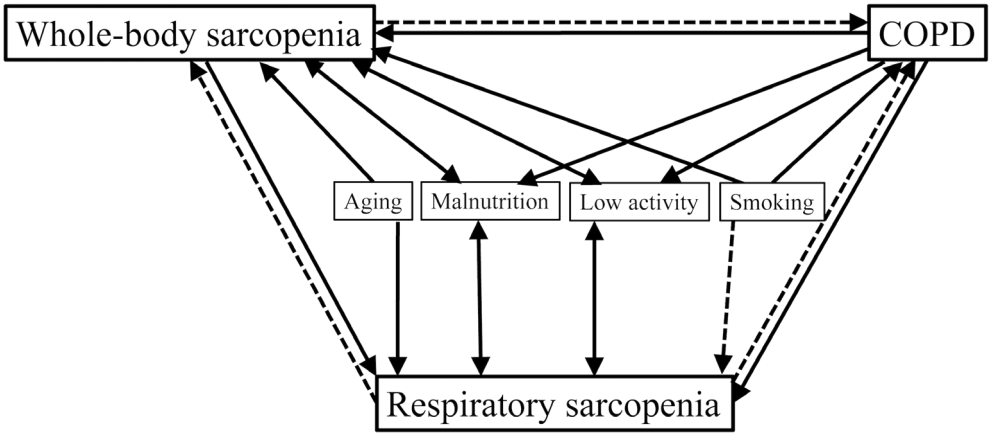


Figure 1 Interaction of whole-body sarcopenia, respiratory sarcopenia and respiratory disease. COPD is shown as a typical example of respiratory disease. COPD causes whole-body sarcopenia and respiratory sarcopenia. Whole-body sarcopenia causes respiratory sarcopenia. Whether respiratory sarcopenia causes whole-body sarcopenia is hypothetical (dashed arrow). Aging causes whole-body sarcopenia and respiratory sarcopenia. COPD causes undernutrition and inactivity. Malnutrition and low activity are associated with whole-body sarcopenia and respiratory sarcopenia. Smoking causes COPD and whole-body sarcopenia. Whether smoking causes, respiratory sarcopenia is hypothetical (dashed arrow). Whole-body and respiratory sarcopenia affect the clinical course and symptoms of COPD. In particular, respiratory sarcopenia may enhance respiratory dysfunction. However, whether whole-body sarcopenia and respiratory sarcopenia directory affect COPD pathologically is hypothetical (dashed arrow). COPD, chronic obstructive pulmonary disease.

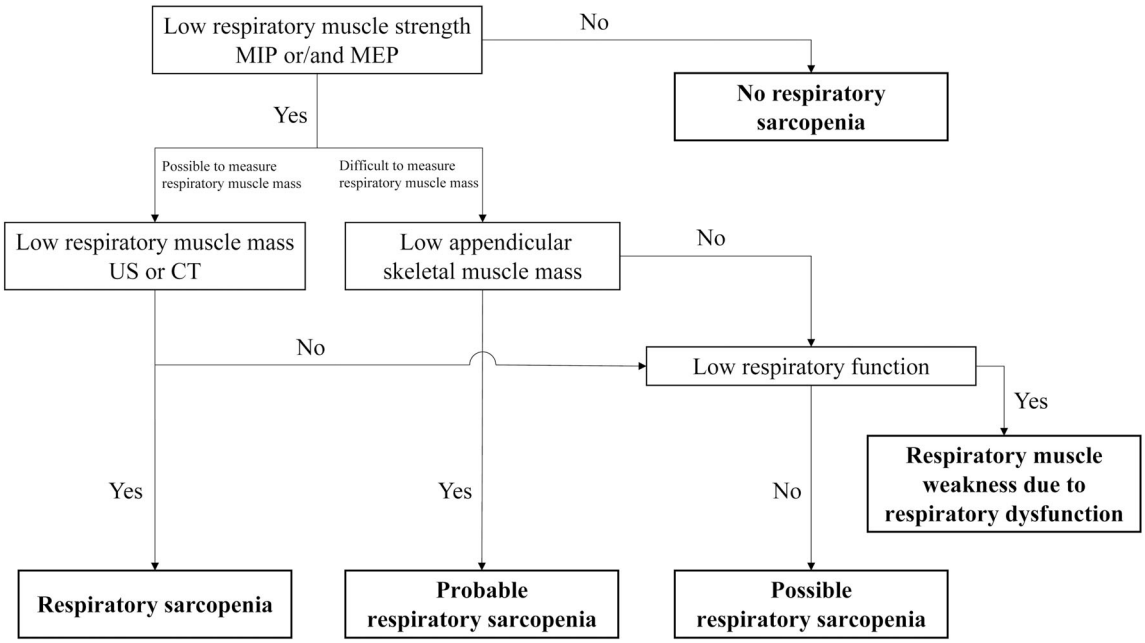


Figure 2 Diagnostic algorithm for respiratory sarcopenia. Diagnosis of "respiratory sarcopenia" is made when low respiratory muscle mass is confirmed in addition to low respiratory muscle strength. If respiratory muscle mass cannot be measured, the diagnosis of "probable respiratory sarcopenia" is made when a low appendicular skeletal muscle mass is confirmed. European Working Group on Sarcopenia in Older People 2, Asian Working Group for Sarcopenia 2019, or other cutoff values can be used to determine the presence of low appendicular skeletal muscle mass. Diagnosis of "possible respiratory sarcopenia" is made when only low respiratory muscle strength is observed without low respiratory function. CT, computed tomography; MEP, maximal expiratory pressure; MIP, maximal inspiratory pressure; US, ultrasound.

Chronic obstructive pulmonary disease-specific diaphragmic dysfunction

In addition to shape changes and muscle mass loss, diaphragmatic muscle fibers are also accompanied by qualitative dysfunction such as impaired force generation and decreased myosin heavy chain content per half of the sarcomere.[66] Moreover, COPD exacerbation causes diaphragmatic dysfunction[29] and a vicious cycle of frequent exacerbations. Decreased paraspinal muscles and

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Table 2 Assessment of respiratory muscle strength and muscle mass for diagnosis of respiratory sarcopenia

Parameter	Measuring and reference values
Muscle strength	1. Respiratory muscle strength is assessed using maximal intraoral pressure, which can measure both inspiratory and expiratory muscle strength. 2. Diagnostic criteria for respiratory sarcopenia developed by the Japanese Association of Rehabilitation Nutrition used only MIP as an indicator of respiratory muscle strength.[47] However, MEP is more associated with whole-body sarcopenia than MIP.[48] Therefore, a decrease in MIP and/or MEP is considered respiratory muscle weakness. 3. Many formulas for calculating reference and normal values for MIP and MEP have been reported, including Enright *et al.*,[14] each using different indices such as age, height, and weight.[49] Evans *et al.*[50] present estimation equations for the reference and lower limit of normal. 4. In Asian populations, it is preferable to use the reference values from the Suzuki *et al.* equation[51] because of the possibility of overestimating or underestimating the MIP and MEP.[49] 5. If evaluation with MIP or MEP is difficult, other respiratory muscle force measures such as PCF, PEFR and SNIP can be used.
Muscle mass	1. Ultrasonography and CT are commonly used to assess respiratory muscle mass. 2. Among the respiratory muscles, the diaphragm, intercostal muscles and accessory respiratory muscles such as the scalene muscles, sternocleidomastoid muscle, pectoralis major muscle and paraspinal muscles can be measured for muscle mass. 3. Lower limit for diaphragm muscle thickness in healthy adults is 0.15 cm[25] and may be used to determine respiratory muscle volume loss. However, there is a wide range of reference values. 4. There are insufficient data to measure quantitatively the respiratory muscle mass including accessory respiratory muscles, and to define a cutoff value for determining low respiratory muscle mass, making it difficult to identify a definitive diagnosis of respiratory sarcopenia. 5. EWGSOP 2[35] and the AWGS 2019[36] for Asian populations are employed to discriminate against low appendicular skeletal muscle mass.

AWGS, Asian Working Group for Sarcopenia; CT, computed tomography; EWGSOP 2, European Working Group on Sarcopenia in Older People 2; MEP, maximal expiratory pressure; MIP, maximal inspiratory pressure; PCF, peak cough flow; PEFR, peak expiratory flow rate; SNIP, sniff nasal inspiratory pressure.

intercostal muscles may also contribute to respiratory muscle dysfunction. Some have proposed twitch mouth pressure (TwPmo) as a functional assessment.[67] MIP and TwPmo by the disease severity are related, and TxPmo may be a convenient measurement index.

Clinical problems

Inspiratory muscle weakness is associated with failure in inhalation therapy, because inhalation therapy, including bronchodilators, is the primary pharmacotherapy for COPD, and various devices are available, including pressurized metered dose inhalers and dry powder inhalers (DPI). Inspiratory flow rate measurement is utilized in clinical practice, particularly because a certain amount of inspiratory flow rate is required to use a DPI.[68] Therefore, respiratory sarcopenia is an important indicator for compounds/devise selection in inhalation therapy. The inspiratory flow rate of <60 L/min suggests a decreased treatment efficacy of DPI.[69]

Pneumonia

Prevalence

Decreased MIP was a risk for the development of pneumonia in a study of 47 inpatients with pneumonia and 35 controls without pneumonia aged ≥70 years.[53] Moreover, a high rate of sarcopenia (53.0%) was present among 637 inpatients in convalescent rehabilitation wards, suggesting a high rate of patients with respiratory sarcopenia.[70]

Mechanisms

Various respiratory diseases and associated conditions, such as susceptibility to infection, risk of exacerbation and underlying inflammation, may lead to respiratory sarcopenia. Moreover, aspiration pneumonia is noteworthy because it is one of the consequences of respiratory sarcopenia due to impaired defense as the "direct cause." In addition, regardless of the underlying disease, aspiration pneumonia is associated with sarcopenic dysphagia.[71]

Decreased coughing, which is directly related to respiratory muscle weakness, is associated with the development of pneumonia. Consequently, pneumonia causes progressive muscle dysfunction, creating a vicious cycle.[72]

Outcomes

Evidence for most critically ill cases of pneumonia can be referred to as patients in the ICU. Decreased iliopsoas muscle assessed using CT, which is an indicator of whole-body sarcopenia in the ICU, was associated with mortality in patients with community-acquired pneumonia.[73] PCF is utilized in neuromuscular diseases.[2] Respiratory sarcopenia is a risk for developing pneumonia both in respiratory diseases and neuromuscular diseases, and PCF is useful in assessing the risk of aspiration pneumonia.

Other respiratory infections

COVID-19

Most reports on COVID-19 are about patients in the ICU. While many patients with COVID-19 underwent chest CT, functional assessment is complex because of aerosol generation. Therefore, a limited number of reports on muscle mass using chest CT images are available.[74] Tdi in patients with severe COVID-19 is associated with reduced contractility[75] and is suggested to be a major cause of long COVID. Respiratory muscle weakness remained after 6 months in severely ill and mildly ill patients who required ICU

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Table 3 Outcomes of respiratory sarcopenia

Category	Outcomes
Physical performance	1. MIP is associated with PCF,[43] while respiratory muscle strength is also related to handgrip strength, lower extremity muscle strength and gait speed.[22,52] 2. When neither whole-body sarcopenia nor low respiratory muscle strength is present, the "robust group"; when only low respiratory muscle strength is present, the "low respiratory muscle strength group"; and when both whole-body sarcopenia and low respiratory muscle strength are present, the "respiratory sarcopenia group," the handgrip strength, knee extension muscle strength, normal gait speed, chair rise and one leg stand time are significantly lower in the respiratory sarcopenia group than in the robust group. The low respiratory muscle strength group had a significantly lower normal gait speed than the robust group.[46]
ADL	1. Independence of ADL was significantly lower in the respiratory sarcopenia group than in the only low respiratory muscle strength group and the robust group.[46] 2. There may be a difference in outcomes between respiratory sarcopenia (respiratory sarcopenia or probable respiratory sarcopenia) and possible respiratory sarcopenia.
Prognosis	1. Risk factors for developing pneumonia in older people included respiratory muscle weakness, decreased trunk muscle mass and malnutrition.[53] 2. Respiratory muscle weakness is associated with an increased risk of mortality in older people.[54] 3. Diaphragm atrophy in ventilator-controlled patients is associated with an increased risk of reintubation and tracheostomy, prolonged duration of ventilatory management and increased respiratory complications.[55] 4. Low MIP is an independent risk factor for long-term mortality in ICU patients requiring ventilation.[56]

Abbreviation: ADL, activities of daily living; MIP, maximal inspiratory pressure; PCF, peak cough flow.

admission; in contrast, few reports evaluated respiratory muscle strength.

Mycobacterial infection. Mycobacterial infections are marked by weight loss and may indicate the difficulty in expelling sputum due to respiratory muscle weakness. In addition, paraspinal muscle mass loss is associated with a poor prognosis.[76]

Lung cancer

Prevalence

Many reports evaluate the thoracic muscle group, paraspinal muscles and iliopsoas muscles using thoracoabdominal CT, routinely taken in clinical practice,[77–79] although few have evaluated respiratory muscle strength. Most reports are only at the preoperative stage and in operable patients, which may have potential selection bias. In a small study of 26 patients with lung cancer, MIP and MEP were reduced by approximately 15–20% compared with 20 healthy people.[80]

Mechanism

It is well known that cancer may associate with cachexia and sarcopenia, and it is hard to distinguish the two. Moreover, caution should be exercised solely based on muscle weakness alone in determining respiratory sarcopenia because of tumor involvement of the phrenic nerve or direct invasion, which causes direct impairment of respiratory muscle function. Furthermore, respiratory muscle weakness can occur after pneumonectomy due to deformity of the chest cage, including the diaphragm, and surgery invasion of respiratory muscles.

Lung cancer and COPD, which share common risks, coexist at a high rate. Respiratory dysfunction related to other respiratory diseases, such as pulmonary fibrosis, also coexists, making it difficult to explain the functional impairment caused by lung cancer alone. Dyspnea on exertion is associated not only with abnormal ventilatory function but also with muscle weakness. Therefore, an appropriate evaluation is required.

Outcome

Impaired activities of daily living or performance status due to muscle weakness may result in difficulty for intensive anticancer treatment and, consequently, lead to a poor prognosis. Some articles reported impaired life expectancy,[79] and one report showed the effect of respiratory muscle training and the occurrence of pulmonary complications up to 2 weeks postoperatively.[81] However, another report showed that short-term postoperative complications did not influence respiratory muscle function.[78] Coexisting COPD may cause respiratory sarcopenia independently and can affect prognosis, and pharmacologic therapy for COPD may improve prognosis.[82]

As no additional clinical effect was proven, specific interventions for respiratory muscle and respiratory muscle training (IMT) have limited recommendations for lung cancer.[83]

Interstitial lung disease and lung transplantation

Prevalence

Reports are limited, and one comparison with neuromuscular disease has been reported.[84] A significant association between MIP and thoracic muscle mass was shown.[85] There is no relationship with hypoxemia.

More than half of the patients have interstitial lung disease in lung transplantation. Lung transplantation does not guarantee disease remission and requires multifaceted and sustained intervention, and there are reports on muscle strength and mass assessment in lung transplant recipients.[86] In muscle mass assessment in lung transplant recipients, there is a significant decrease in muscle mass before lung transplantation, with an average decrease in the paraspinal muscle of more than 20% compared with healthy donors.[87]

Mechanism

The lung parenchyma becomes less compliant, and the respiratory muscular dysfunction causes impaired lung expansion, resulting in restricted ventilatory failure. Unlike COPD, Laplace's law favors diaphragmic pressure, but the decreased lung compliance and increased respiratory workload are associated with dyspnea on

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exertion and exercise intolerance. Dyspnea and exercise intolerance are associated with decreased MIP and changes in Tdi.[88] Inspiratory capacity is essential for dyspnea on exertion, similar to COPD, suggesting a disease non-specific association.[89]

Outcome

As a preoperative CT-based assessment of paraspinal muscles before lung transplantation, a decrease in muscle mass and quality affects the course of postoperative muscle mass recovery and is associated with prognosis and chronic lung allograft dysfunction.[34] Conversely, MIP improves after lung transplantation from a decreased preoperative status and is consequently associated with improved lung volume, following improvement of exercise tolerance.[86] Therefore, MIP is an important determinant of exercise tolerance in addition to a lower extremity strength after lung transplantation.

ICU

Prevalence

Trunk muscle mass was noted as an established prognostic factor in patients with various diseases, surgeries, and trauma, although prevalence data are scarce. ICU-acquired weakness is a sequela in critically ill patients and is considered systemic muscle weakness. ICU-acquired weakness, referred to as post-intensive care syndrome, also impairs respiratory muscles, resulting in both whole-body and respiratory sarcopenia.

Mechanism

Systemic muscle dysfunction is caused by complex factors such as disuse atrophy, myopathy due to underlying diseases and systemic inflammation related to critical illnesses that require ICU management, and myopathy caused by drugs such as sedatives and anesthetics. Moreover, functional impairment of individual muscle fibers plays an essential role in the pathogenesis of muscle degeneration due to oxidative stress and impaired blood flow to the diaphragm.

Ventilatory management induces diaphragmatic atrophy, which affects clinical outcomes.[55] In addition, severe status and prolonged ventilatory management caused interrelated ICU-induced respiratory sarcopenia.

Outcome

Diaphragm function assessment using ultrasonography is used to evaluate respiratory muscle function to predict reintubation[8] and extubation feasibility. Therefore, these parameters are a critical clinical indicator in the ICU. Respiratory sarcopenia is associated with reintubation rates[90] and is a prognostic factor in life-threatening pneumonia.[73]

Therefore, quantitative thoracoabdominal CT and other imaging techniques are used as outcome measures. In addition, ultrasonography is used to evaluate diaphragm function at each point in the course of the disease and to evaluate changes over time.

Prevention and treatment of respiratory sarcopenia

As defined here, no previous studies have examined the efficacy of preventive and therapeutic interventions for respiratory sarcopenia. As with whole-body sarcopenia, effective pharmacologic interventions are unavailable for respiratory sarcopenia. Here, we review respiratory muscle intervention studies and present exercise and nutritional interventions.

Exercise intervention

Respiratory muscle training, including IMT, is the most promising intervention for respiratory sarcopenia, improving respiratory muscle strength, respiratory function, physical function and health-related quality of life (HRQoL). IMT in healthy older adults increased MIP and Tdi.[91] Expiratory muscle training increases MEP. However, the effects on coughing power and pulmonary function are unclear.[92] Respiratory muscle training increased MIP, MEP, and vital capacity in smokers and non-smokers.[93] In older patients with whole-body sarcopenia, whole-body strength training significantly improved maximum expiratory flow, appendicular skeletal muscle mass and handgrip strength compared with controls.[94] The effects of respiratory muscle training on physical function and QoL in healthy older adults are unclear and require further study.[91]

In patients with COPD, IMT improved MIP, forced expiratory volume in 1-s, exercise tolerance, dyspnea and HRQoL.[11,95] In patients with heart failure, MIP, exercise tolerance and HRQoL were improved.[96] In patients with stroke, respiratory muscle training increases MIP and MEP, forced expiratory volume in 1-s, FVC and PEFR increases 6MWD and improves swallowing function.[97,98] In addition, IMT significantly led to a shorter duration of mechanical ventilation in patients receiving mechanical ventilation compared with conventional physical therapy alone.[99]

In patients undergoing cardiac or abdominal surgery, preoperative IMT was associated with lower postoperative pulmonary complications and shorter hospital stays.[83] In addition, in stroke patients, respiratory complications were reduced.[97,98] Thus, while respiratory muscle training prevents complications, the effect of an exercise intervention on preventing the development of respiratory sarcopenia and improving long-term prognosis is not apparent.

Nutritional intervention

Nutritional interventions have beneficial effects on skeletal muscle mass, exercise capacity and respiratory muscle strength. Nutritional supplementation in stable patients with COPD improves the weight, skeletal muscle mass, 6MWD and HRQoL. In addition, nutritional supplementation for malnourished patients with COPD significantly improved MIP and MEP.[100] A leucine-enhanced nutritional intervention in older patients significantly decreased walking speed and MEP in the placebo group compared with the intervention group,[101] suggesting that nutritional intervention may prevent respiratory sarcopenia.

Combined interventions

Combining both exercise and nutritional interventions is important to improve outcomes in disabled older adults with malnutrition and whole-body sarcopenia and may be helpful for respiratory sarcopenia. For example, in patients with COPD, nutritional support during pulmonary rehabilitation significantly improved quadriceps strength and MIP.[102] In addition, nutritional support during high-intensity exercise training in patients with COPD with reduced skeletal muscle mass increased body weight, physical activity and inspiratory muscle strength.[103]

Future perspective

Future studies on the prevalence and prognosis of respiratory sarcopenia are required.[104] The reliability and validity of diagnostic criteria for respiratory sarcopenia should be verified. Validations

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of the usefulness of prevention and treatment of respiratory sarcopenia are required.

Concerning respiratory muscle mass, standardization of measurement methods and the development of cutoff values and minimal clinically important differences are required. The development of methods to measure respiratory muscle mass easily is also important. MEP and MIP should be measured more in clinical practice. Studies examining changes in respiratory muscle mass and strength over time are expected. Developing screening methods for respiratory sarcopenia and diagnostic methods in settings where testing equipment is limited is required. Moreover, studies examining the relationship between respiratory sarcopenia, whole-body sarcopenia and sarcopenic dysphagia are expected.

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Data availability statement

Research data are not shared.

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